JunHyun Sung
Department of Architectural Engineering
Pusan National University
63 Busandaehak-ro, Geumjeong-gu, Busan 46241, Republic of Korea
Email: tobunda@pusan.ac.kr

1. RESEARCH INTERESTS

- Computer Vision and Remote Sensing
- Human-Centered AI and Human-Machine Collaborations
- AI-driven Construction Safety Monitoring
- Vision-Language Models and Generative AI

2. EDUCATION

Pusan National University (Busan, Republic of Korea) ..... Sep. 2025 – Present
- *Master of Science in Architectural Engineering* (Advisor: JungHo Jeon)
- Thesis: An rPPG-Based Approach to Contactless Stress Detection and Classification for Construction Sites

Pusan National University (Busan, Republic of Korea) .....Mar. 2018 – Aug. 2025
- *Bachelor of Science in Architectural Engineering*

3. RESEARCH EXPERIENCE

3.1 Research Projects

Development of Construction Safety Management Agent via Human Foundation Model .....Mar. 2026 – Present
supported by National Research Foundation of Korea (NRF)
- Role: Graduate Research Assistant, Pusan National University

Development of Safety Management Platform
using Generative AI-based Data Augmentation Approach.....Mar. 2025 – Present
supported by National Research Foundation of Korea (NRF)
- Role: Graduate Research Assistant, Pusan National University

Development of the BIM-based Decision Support System
for Green Building Planning with Each Design Phase .....Jun. 2024 – Feb. 2026
supported by National Research Foundation of Korea (NRF)
- Role: Graduate Research Assistant, Pusan National University

3.2 Publications (1 published and 1 accepted)

[01] **Sung, J.**, Lee, K., Kim, N., Choi, B., & Jeon, J. (2026). "Contactless Detection and Classification of Workers' Stress via remote Photoplethysmography (rPPG)". *Journal of Computing in Civil Engineering*. ***(Accepted)***

[02] Shin, D., Lee, J., **Sung, J**., & Yoon, S. (2024). "Effects of uncertainty Parameters Occurring in The Design and Construction Process on RC Structural Performance". *Journal of The Earthquake Engineering Society of Korea*, 28(6), 305-314. http://doi.org/10.5000/EESK.2024.28.6.305

3.3 Conferences (4 published)

[05] Jeong, K., **Sung, J**., Jung, D., Lee, K., & Jeon, J. (2026). A Visual Question Answering Framework for Fall-related Safety Compliance Checking in Construction Sites. In *Computing in Civil Engineering 2026*.

[04] Jeong, K., **Sung, J**., Kang, G., Park, G., Park, J., & Jeon, J. (2025). "Construction Safety Compliance Checking using VQA Models: Context-Specific Performance Analysis". In *Proceedings of KICEM Annual Conference 2025* (pp. 45–46). (***Best Paper Award***)

[03] **Sung, J**., Jeong, K., Lee, Y., Kang, G., Park, H., & Jeon, J. (2025). "Multi-class Classification of Stress Levels of Construction Workers Using Remote Photoplethysmography (rPPG)". In *Proceedings of the Korea Institute of Building Construction Annual Conference* (Vol. 25, No. 2, pp. 71–72). (***Academic Excellence Award***)

[02] **Sung, J**., Jo, W., Lim, K., Cho, S., Yoon, S., & Jeon, J. (2025). "A Feasibility Study on Stress Detection of Construction Workers Using rPPG Signals". In *Proceedings of the Architectural Institute of Korea Conference* (Vol. 45, No. 2, pp. 887–890). (***Best Paper Presentation Award***)

[01] Lee, Y., Kang, G., Jo, W., **Sung, J.**, Park, H., & Jeon, J. (2025). "Development of AI-generated images quality assessment framework for construction safety". In *Proceedings of the Architectural Institute of Korea Conference* (Vol. 45, No. 1, pp. 669–670).

4. INDUSTRY EXPERIENCE

Safety Assistant (Busan, Republic of Korea) ..... Dec. 2018 – Feb. 2019

- LCT Construction Project (POSCO E&C)
- Main Tasks: On-site Safety Support

5. HONORS AND AWARDS

Best Paper Award, Korea Institute of Construction Engineering and Management ..... Nov. 2025
Academic Excellence Award, Korea Institute of Building Construction ..... Nov. 2025
Best Paper Presentation Award, Architectural Institute of Korea. ..... Oct. 2025
Merit Scholarship, Pusan National University ..... Mar. 2018

6. LEADERSHIP AND COMMUNITY SERVICES

Republic of Korea Army ..... Apr. 2019 – Nov. 2020

- Squad Leader (Aug. 2020 – Nov. 2020)

7. COMPETENCES

Programming Languages: Python, MATLAB, C, C++, and R
Engineering Software & Tools: PyTorch, AutoCAD, SketchUp, and Twinmotion
Languages:

- English: TOEFL iBT 98 (R: 26, L: 25, S:23, W: 24)
- Korean (Native)

Other: MS Office